Stereo Attention Cross-Decoupling Fusion-Guided Federated Neural Learning for Hyperspectral Image Classification

Weiwei Cai, *Member, IEEE*, Ming Gao, *Member, IEEE*, Yao Ding, Xin Ning, *Senior Member, IEEE*, Xiao Bai, *Senior Member, IEEE*, and Pengjiang Qian, *Senior Member, IEEE*

***Abstract*— Federated learning is a promising solution in several industries for cotraining models among distributed clients via centralized servers without leaving private user data on the devices. Thus, federated learning can be seen as a stimulus for the edge computing paradigm as it supports collaborative learning and model optimization. In view of the strict requirements for data security and system reliability of hyperspectral classification techniques for surveillance, aerospace, and military missions, this article proposes a novel stereo attention cross-decoupling fusion (CDF)-guided federated neural learning algorithm for hyperspectral image classification, which first trains client devices using a scalable federated learning approach consisting of master server, secure aggregator and edge client devices of a certain size. The distributed devices train local models of the neural network for classifying hyperspectral images and send them to the secure aggregator, which aggregates the local models using a weighted averaging strategy and sends them to the master server for iteration. In addition, the stereo attention CDF module is used to mine the multidimensional spatial details of the hyperspectral images, specifically by first extracting the most discriminative features from different directions (horizontal, vertical, and spatial) using the attention mechanism and then using the decoupling fusion strategy to classify the original feature map into three levels: significant, minor, and redundant, and use them to model the multidimensional spatial relationships, thus strengthening the capability to represent features. Extensive experiments on several public datasets have shown that the proposed method provides competitive performance, and more importantly, is effective in enhancing privacy and reliability for hyperspectral image classification.**

***Index Terms*— Edge computing, feature fusion, federated learning, hyperspectral image classification, neural networks.**

Manuscript received 28 May 2023; revised 20 August 2023; accepted 27 August 2023. Date of publication 27 September 2023; date of current version 23 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62172192, Grant U20A20228, and Grant 62171203; in part by the 2018 Six Talent Peaks Project of Jiangsu Province under Grant XYDXX-127; and in part by the Science and Technology Demonstration Project of Social Development of Jiangsu Province under Grant BE2019631. *(Corresponding author: Pengjiang Qian.)*

Weiwei Cai, Ming Gao, and Pengjiang Qian are with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi 214122, China (e-mail: vivitsai@ieee.org; aarongaoming@ieee.org; qianpjiang@jiangnan.edu.cn).

Yao Ding is with the Xi'an Research Institute of High Technology, Xi'an 710025, China (e-mail: dingyao.88@outlook.com).

Xin Ning is with the Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China.

Xiao Bai is with the School of Computer Science and Engineering, Beihang University, Beijing 100191, China (e-mail: baixiao@buaa.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3320044

I. Introduction

THE exponential growth of IoT devices and computing power has resulted in unprecedented amounts of data, which will continue to increase as 5G networks become widespread and the number of mobile devices that are accessed increases exponentially [1]. In the past, both cloud and artificial intelligence technologies attempted to accelerate innovation by deriving actionable knowledge from data [2]; however, the volume and complexity of data generated by connected devices are unprecedented and already exceed the capacity of existing networks and infrastructure [3]. Sending all device-generated data to a centralized data center or the cloud can result in bandwidth and latency problems. Edge computing provides a more efficient alternative by processing and analyzing data nearer to its origin. Because the data is not transferred over the network to the cloud or data center for processing, latency is drastically reduced [4].

Edge computing is an emerging computing paradigm that refers to a set of networks and devices at or near the user [5]. Edge computing involves processing data closer to where it is generated, at a faster rate and with greater capacity, resulting in greater action-oriented outcomes in real time. It offers distinct advantages over traditional models that concentrate computing power in local data centers [6]. Placing computing at the edge can enhance the management and use of physical assets and enable the creation of new interactive human experiences [7]. Autonomous vehicles [8], remote sensing [9], [10], smart device data [11], and automated retail [12] are a few examples of edge use cases.

Edge computing combined with deep neural networks is a promising technology [13] and is widely used for a variety of applications; however, in traditional deep neural network architectures based on edge computing, data producers must often send data and share it with third parties, edges, or cloud servers to train their models. Because of high bandwidth requirements, legalization, and privacy risks, such structures are typically unfeasible. The concept of federated learning [14] has recently emerged as a promising solution to alleviate the problems of unnecessary bandwidth loss, data privacy, and legitimization. Federated learning allows for cotraining of models across distributed clients (e.g., mobile phones, automobiles, hospitals, etc.) via centralized servers while maintaining data localization [15]; therefore, federated learning can be seen as a positive stimulus in the edge computing paradigm as it

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

supports collaborative learning [16] and model optimization. Specifically, the use of federated learning for model training on mobile edge networks offers a number of advantages over traditional cloud-centric training methods [17]. By using federated learning provides for a more efficient use of network bandwidth [18], as less data is transported to the cloud as compared to traditional methods. For instance, instead of transmitting raw data for processing, devices involved in training only transmit updated model parameters for aggregation [19], thereby considerably decreasing the cost of data communication and lessening the strain on the backbone network. Assuming federated learning participants and servers are in a nonmalicious state [20], the usage of federated learning can improve user privacy [21] and lessen the likelihood of eavesdropping due to the fact that the user's raw data does not need to be forwarded to the cloud. Additionally, as anonymity is better safeguarded, more people are ready to participate in collaborative model training, enabling the development of superior neural network models. Using federated learning, deep neural network models [22] can be consistently trained and updated with respect to latency. Most decisions in the edge domain can be made in real time at the edge node or end device; thus, the latency is much less than generating choices in the cloud and then communicating them to end devices, which is crucial for airborne HSI classification tasks [23], [24].

Hyperspectral images have the advantages of narrow spectral range, continuous bands, approximate spectral information of ground features, and strong feature recognition ability; however, the high dimensionality and rich band information of hyperspectral images generate a large amount of redundant data, which brings a huge computational burden. Although some dimensionality reduction methods [25] greatly alleviate this limitation, HSI recognition tasks are often limited by the computational power of airborne and spaceborne computers [26], so federated learning techniques in edge computing are expected to provide a solution to this limitation.

This article, therefore, proposes a novel stereo attention cross-decoupling fusion (CDF) guided federated neural learning algorithm for HSI classification. The proposed deep neural network is first trained using a scalable mobile device-based federated learning approach at scale, which consists of a master server, a secure aggregator, and a certain size of edge node devices. The distributed mobile device trains the local model of the neural network for classifying HSIs and sends it to the secure aggregator, which aggregates the local model using a weighted averaging strategy and sends them to the master server for iteration. Federated neural learning methods can significantly improve the privacy and reliability of HSI classification, especially for astronautical and military missions, and to the best of our knowledge this article is the first time that federated learning techniques are used for HSI classification. In addition, in the proposed algorithm, a stereo attention CDF module is used to mine the multidimensional spatial details of HSIs, specifically first extracting the most discriminative features from three directions (horizontal, vertical, and spatial) step by step using an attention mechanism, followed by a decoupling fusion strategy to classify the original feature map into three levels: significant, minor, and redundant, and use them to model multidimensional spatial relationships, thus enhancing the feature representation capability. Although the division of the dataset into multiple in the federated learning task, extensive experiments on multiple open-source baseline datasets show that the proposed algorithm still provides competitive performance, especially in terms of unique privacy and reliability compared to many existing methods. The main contributions of this article are as follows.

1) This article provides an improved federated neural learning method for the hyperspectral classification task. Specifically, when distributing the sample data to the client devices, the independence of each sample data is guaranteed, while letting them have the same feature space, which improves the privacy and reliability of the data of the classification system.
2) In this article, we propose a novel stereo attention CDF-guided federated neural learning model, which can use the multidimensional spatial details of hyperspectral images to perceive decision boundaries between important and minor semantics by using an attention mechanism to progressively extract multidirectional semantic features from three directions (horizontal, vertical, and spatial).
3) In this article, we propose a novel feature decoupling rating module that rates features into three types of features, i.e., important, minor, and redundant, and feature selection is implemented by a threshold strategy to extract more discriminative features.
4) Extensive test results on four well-known HSI datasets indicate that the proposed algorithm achieves competitive results in both time consumption and reliability, especially in the small sample case.

The remainder of the article is organized as follows. Section II describes the related work. Section III discusses the proposed algorithm in the context of edge computing and federated learning, with a special emphasis on stereo attention CDF guided federated neural learning model for classification of HSIs. Section IV presents experimental results and analyzes the test results. Section V gives conclusions and indicates direction for future work.

II. Related Work

A key set of reasons for transitioning from a traditional cloud-based model to an edge computing platform, in addition to low latency and high bandwidth, is that the edge also provides greater security. For example, by sending data to the edge device, attackers will have less time to launch an attack because the latency of the edge device is lower than that of the cloud. Attacks such as DDoS, which typically cripple cloud-based systems, furthermore become largely harmless in an edge computing environment because compromised edge devices can be removed without affecting the overall performance of the network. As a result, edge networks are more reliable because they have no single point of failure. As mentioned earlier, edge networks are easier to scale because of their smaller size [27]. In fact, a scalability strategy that scales outward rather than upward is a very attractive way to achieve high performance at minimal cost; moreover, some of these edge devices or edge data centers may not even need to be built from scratch, and different stakeholders can collaborate to share current IoT edge device resources [28]. Following

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

are the key principles upon which edge computing systems are based.

1) *Mobility:* For applications such as self-driving cars, edge devices must accommodate end users who are always in motion without compromising latency or bandwidth. Some techniques solve this issue by placing roadside edge devices.
2) *Proximity:* To guarantee low latency, the edge device must be as close to the end user as possible. This may involve conducting computation directly on the edge device or investing in a local edge computing data center in close proximity to the end user. For ubiquitous edge computing, the network must have a vast reach.

The correct deployment of nodes in a framework for edge computing is, therefore, essential for guaranteeing the best possible user experience. Clearly, dense distribution is desired, but economic constraints must be taken into account.

Federated learning is typically divided into horizontal federated learning, vertical federated learning, and federated transfer learning based on the distribution of data in the feature and sample space.

1) Horizontal federated learning, or sample-based federated learning, is introduced in scenarios where the datasets share the same feature space but have different samples. For example, two user groups in different regions may have a very small cross-set of users due to their very different regions; however, if their attributes are very similar, the feature space is the same. Thus horizontal federated learning can be summarized by the following expression:

$$\begin{cases} X_i = X_j \\ Y_i = Y_j \quad \forall D_i, D_j, \quad i \neq j \\ I_i \neq I_j \end{cases} \tag{1}$$

where D_i and D_j denote the data held by each data owner i and j, feature X, and label Y. Horizontal federated learning is generally only available to servers to jeopardize the privacy of data participants.

Vertical federated learning for the case where two datasets share the same sample ID space but have different feature spaces. For example, consider two different companies in the same city; one is a bank, and the other is an e-commerce company. Their user sets may contain most of the residents in the area, so the crossover of their user spaces is large; however, since the bank records the users' income and expenditure behavior and credit ratings, and the e-commerce retains the users' browsing and purchasing history, their feature spaces are very different. Vertical federated learning can be summarized by the following expression:

$$\begin{cases} X_i \neq X_j \\ Y_i \neq Y_j \quad \forall D_i, D_j, \quad i \neq j \\ I_i = I_j \end{cases} \tag{2}$$

where D_i and D_j denote the data held by each data owner i and j, feature X, and label Y.

Federated transfer learning for circumstances in which the sample sizes and feature spaces of two datasets are distinct. It uses a small number of similar samples to discover a common representation across the two feature spaces, which is then used to get predictions for samples with only one-sided features. Federated transfer learning is a significant addition to existing federated learning systems and may be summed up as follows:

$$\begin{cases} X_i \neq X_j \\ Y_i \neq Y_j \quad \forall D_i, D_j, \quad i \neq j \\ I_i \neq I_j \end{cases} \tag{3}$$

where D_i and D_j denote the data held by each data owner i and j, feature X, and label Y.

In this article, the proposed federated learning system architecture adopts a horizontal model since the same hyperspectral data is distributed to different mobile edge devices for training.

III. Methodology

The structure of the proposed stereo attention CDF guided federated neural learning model for HSI classification is shown in Fig. 1. Specifically, client devices such as drones and spaceborne and airborne devices request to connect to the server and send the collected spectroscopic data to the server. Some requests will fail due to security risks or network anomalies. After establishing a successful connection with the server, the server sends the stored aggregation model and configuration file to the client device, and at the same time, sends the received hyperspectral data divided into K parts according to labeling categories to the client device. Then, the client trains the model with a neural network guided by the stereo attention CDF model. After model fitting, the client device sends the trained HSI model to a secure aggregator, which aggregates the received model according to the aggregation operator and sends it to the server for persistent storage. Since the K data distributed by the server have the same feature space distribution, a single client device can actually accomplish the task of model training. This means that the proposed algorithm has high security, even if $K - 1$ client devices are lost, the classification of HSIs can be completed. In addition, the federated neural network can also greatly accelerate the reasoning efficiency of the model; therefore, the proposed algorithm has positive significance in the military and industries with high-security requirements. Next, this article will describe the algorithm in detail.

A. Federated Neural Learning

The federated learning model described in this article for HSI classification is called horizontal federated learning because it is based on partitioning data samples. Horizontal federated learning is also referred to as sample-partitioned federated learning or example-partitioned federated learning and is applicable to all participants in federated learning. Similar to the scenario where data is horizontally partitioned in a table view, the datasets may have the same feature space and different sample spaces.

The term "horizontal partitioning" is commonly employed in conventional database table display settings. For instance, entries in a table are horizontally separated into several groups based on rows, and each row has comprehensive data attributes. Generally, the industry thinks that all participants in a horizontal federated learning system are trustworthy, and the target to be guarded against is a trustworthy but nosy aggregate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

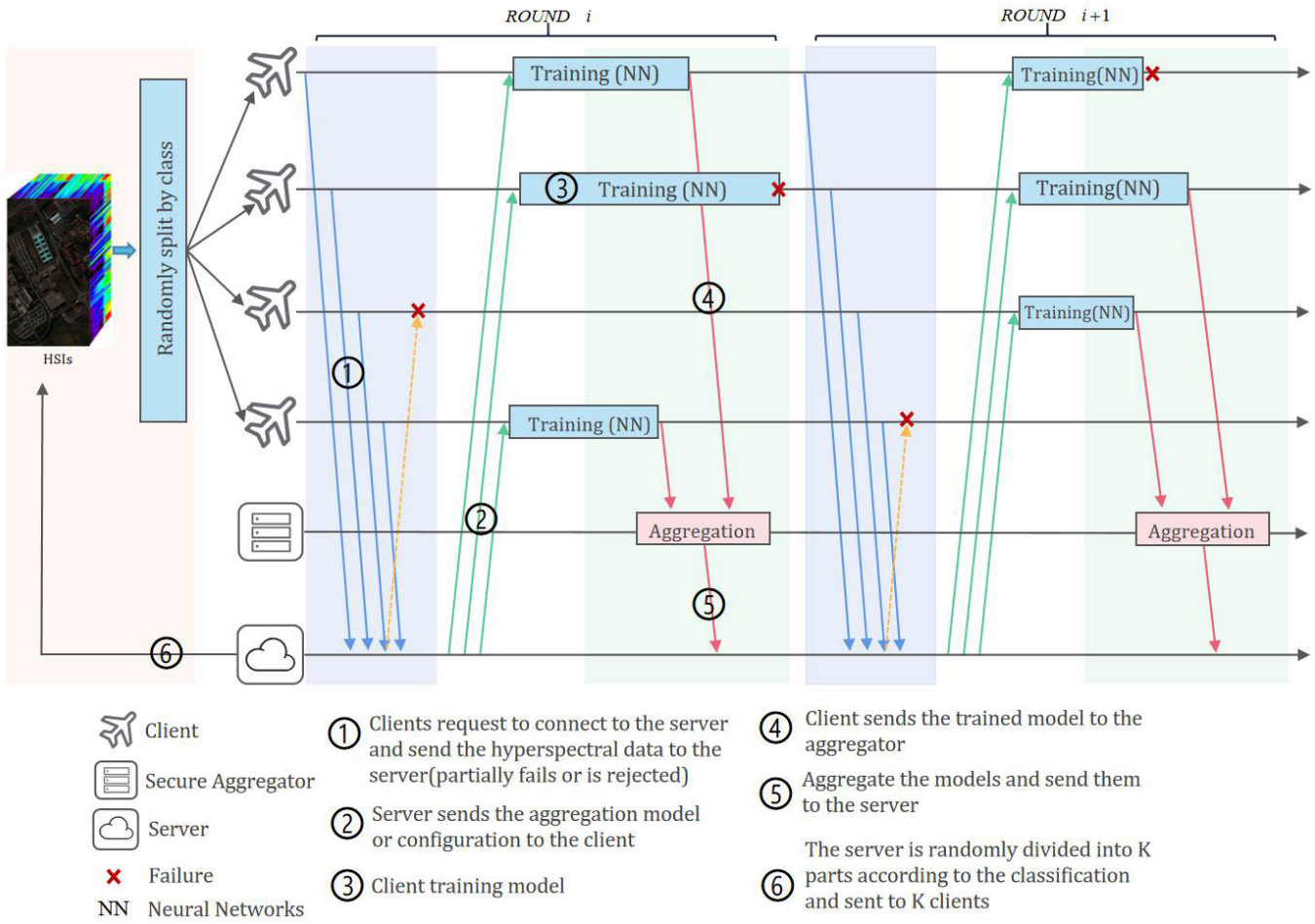


Fig. 1. Schematic of hyperspectral image classification based on federal neural network.

server. In other words, it is commonly assumed that only the server poses a privacy risk to data participants.

This article adopts the master–slave architecture mode, and there are K clients $\{C_1, C_2, \ldots, C_K\}$ with the same data structure, each of which has a dataset (X_K, Y_K), where each X_K has the same feature space. The stereo attention CDF guided federated neural learning model for HSI classification is trained collaboratively with the help of servers (master and aggregation servers). In this article, X_K, Y_K, and I_K are used to represent the feature space, label space, and training sample ID space of the client C_K, respectively. Since we divide the HSIs into K parts and distribute them to K clients, each label space Y_K is the same, and the sample space X_K is different but has a common feature space

$$Y_1 = Y_2 = \cdots = Y_K \tag{4}$$

$$\bigcap_{k=1,2,\ldots,K} X_k^{\text{feature}} \neq \emptyset \tag{5}$$

$$\bigcap_{k=1,2,\ldots,K} X_k^{\text{sample}} = \emptyset. \tag{6}$$

Different from the traditional horizontal federated learning mechanism, the purpose of this work is not to improve the performance of the neural network model by increasing the number of data samples but to divide the dataset into individual clients for training to enhance reliability. It is worth noting that doing so does not affect the classification performance of HSIs, as there is a large body of existing literature showing that deep neural network models require only a small amount of training to achieve surprising classification accuracy. Fig. 2 shows a schematic of the dataset structure and feature space distribution under each client.

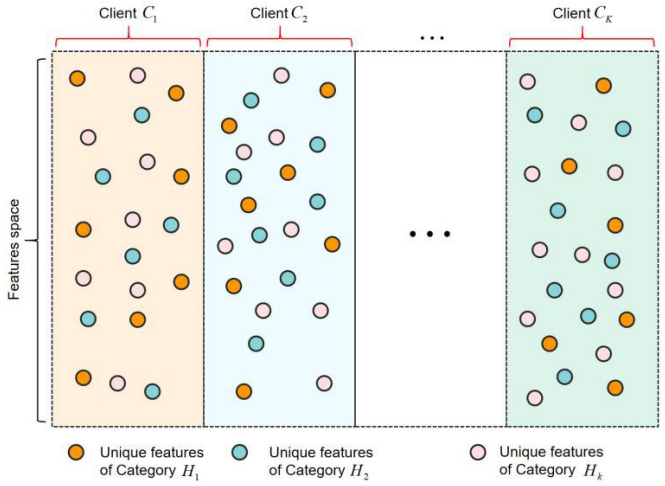


Fig. 2. View of dataset structure and features space in the clients scenario.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

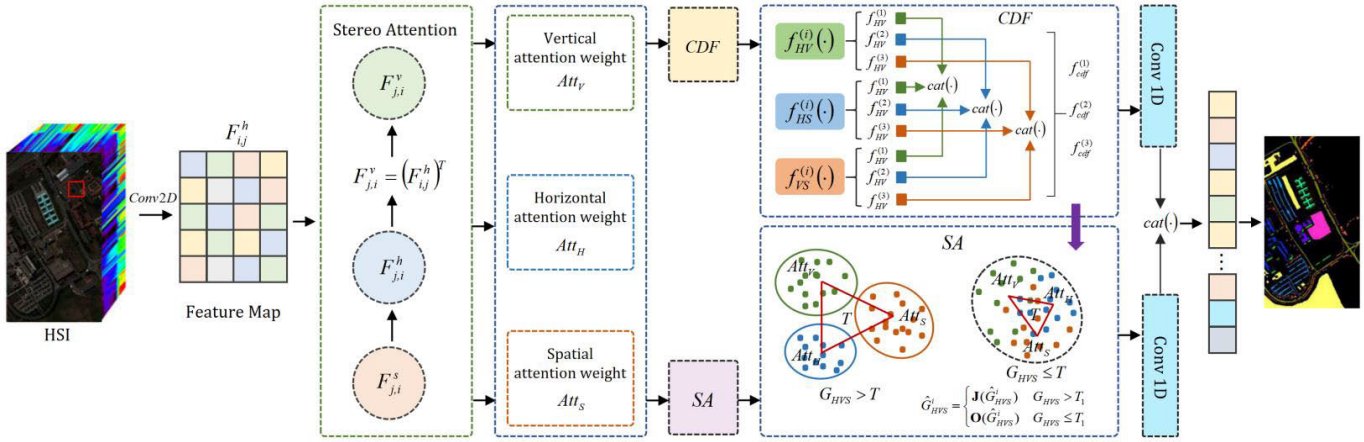


Fig. 3. Schematic of stereo attention cross-decoupling fusion mode.

As shown in Fig. 2, the datasets received by each client device have the same feature space distribution, which means that as long as the number of clients is not particularly large, a single client device can independently complete the training task. In this case, ordinary horizontal federated learning only uses the common features $\{X_K^c\}_k$ to jointly train the federated neural network, thus, the independent features $\{X_K^u\}_k$ are not used. The goal of this article is to improve the reliability and inference efficiency of the horizontal federated learning model by randomly dividing the dataset into K clients by category labels. Please note that the method sought has the following characteristics: high reliability, that is, the dataset is distributed to K clients by category labels; each client device can independently accomplish the classification task of HSIs, and the training efficiency of the neural network can be further improved.

In addition, secure aggregation, which is a secure multiparty computation protocol, uses encryption to make individual device updates uncheckable by the server and only displays the sum after a sufficient number of updates have been received. In this article, we are deploying secure aggregation as a privacy enhancement to federated learning, which prevents data leakage by ensuring that individual device updates remain encrypted even in memory. Let the initial weight of the security aggregator be w_0 and for each round of federated learning, K successfully connected clients are selected to compute the weight updates

$$\overline{w} = \sum_k \Delta^k \tag{7}$$

$$\overline{n}_t = \sum_k n^k \tag{8}$$

$$\Delta_t = \frac{\Delta_t^k}{\overline{n}_t} \tag{9}$$

$$w_{t+1} \leftarrow w_t + \Delta_t \tag{10}$$

where $\overline{w}_t$ denotes the product of K client weight updates, $\overline{n}_t$ denotes the product of weights, and Δ_t denotes the average update. Let the client divide the dataset into B batches, then

$$w \leftarrow w - \eta \nabla \ell(w; b) \tag{11}$$

$$\Delta \leftarrow n(w - w_{\text{init}}) \tag{12}$$

where Δ denotes the weight of the kth client update.

B. Stereo Attention Cross-Decoupling Fusion Model

The proposed SACDF model is composed primarily of a CDF module and a semantic aware (SA) module. The SACDF model first captures the shallow features of HSI by convolution operations and then computes the attention weight matrices Att_H, Att_V, and Att_S from different directions, horizontal, vertical, and spatial, using a three-direction attention mechanism. By combining them, three new attention matrices Att_{HV}, Att_{HS}, and Att_{VS} are obtained. Then, based on the double-threshold strategy, CDF divides these three combined attention weights into three levels: important, minor, and redundant. Then, by fusing the feature information of corresponding levels, three new feature matrices f_1, f_2, and f_3 are also obtained. Meanwhile, SA extracts the optimal shallow global semantic feature f_g of f_1, and the optimal deep local semantic feature f_d of the minor feature f_2. Taking into account the fact that redundant features contain less useful semantic information. In this work, redundant features are compressed into 1-D dense vectors to reconstruct their relationship. Fig. 3 presents the schematic of the proposed SACDF.

C. Stereo Attention Mechanism

In this section, we propose a stereo attention mechanism that assigns weights to each feature from vertical, horizontal, and spatial directions to extract stereo attention features. Let the feature map extracted from the previous convolutional layer be F_{ij}, where h and w represent the height and width of the feature map, and F_{ij} is input to the horizontal attention unit to calculate the weight in the horizontal direction, and its calculation formula can be defined as follows:

$$\text{Att}_H = \frac{e^{(W_h F_{i,j} + b_h)}}{\sum_{i,j} e^{(W_h F_{i,j} + b_h)}} \tag{13}$$

where W_h and b_h are the parameters of the fully connected layer and Att_H is the coefficient along the horizontal attention.

The weight in the vertical direction is calculated by inverting the feature matrix, and the process is expressed by the following equation:

$$F_{j,i}^v = \left(F_{i,j}\right)^T \tag{14}$$

where $F_{j,i}^v$ indicates the feature map that has been transposed; therefore, we can obtain attention weights in the vertical

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

direction in the same way, whose formula is declared as follows:

$$\mathrm{Att}_V = \frac{e^{(W_v F_{j,i} + b_v)}}{\sum_{j,i} e^{(W_v F_{j,i} + b_v)}}. \tag{15}$$

For spatial direction attention, it focuses on the distribution of weights between feature maps to further obtain spatial features. Assuming that the features of the input spatial direction attention module are $(F^s_{i,j})^k \in R^{H\times W\times k}$, $(F^s_{i,j}) = [(F^s_{i,j})^1, (F^s_{i,j})^2, \ldots, (F^s_{i,j})^k]$, and k is the feature map index, and the weight in the spatial direction is expressed by the following equation:

$$\mathrm{Att}_S = \frac{e^{(W_s F_{i,j} + b_s)}}{\sum_{i,j} e^{(W_s F_{i,j} + b_s)}} \tag{16}$$

where Att_S denotes the weight coefficient along the spatial attention.

D. Cross-Decoupling Fusion Module

The CDF module consists of two parts: dual-threshold screening and feature fusion. In double threshold screening, the attention weight matrices Att_H, Att_V, and Att_S are categorized into important, minor, and redundant levels according to two thresholds T_1 and T_2. Then, by reconstructing Att_H, Att_V, and Att_S in pairs, three new weight matrices Att_{HV}, Att_{HS}, and Att_{VS} can be obtained, and then the mean values of them are taken to obtain $((\mathrm{Att}_{HV} + \mathrm{Att}_{HS})/2)$, $((\mathrm{Att}_{HS} + \mathrm{Att}_{VS})/2)$, and $((\mathrm{Att}_{HV} + \mathrm{Att}_{VS})/2)$, which are compared with the thresholds T_1 and T_2 one by one for comparison. The motivation for taking the mean value of the three-directional attention weights is that it is more stable to compute the mean value of any two sets of weights when ranking the feature points. If its value is greater than T_1, it means that the feature point contains important semantic information, and we use the all-ones matrix $\mathbf{J}$ to enhance its features. Similarly, if the value is less than T_2, it indicates that the feature point is redundant information, and the all-zeros matrix $\mathbf{O}$ is used to penalize it, and the computational equation of CDF can be defined as follows:

$$a^{(1)}_{HV} = \begin{cases} \mathbf{J}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} > T_1 \\ \mathbf{O}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} \le T_2 \end{cases}$$

$$a^{(1)}_{HS} = \begin{cases} \mathbf{J}, & \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} > T_1 \\ \mathbf{O}, & \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} \le T_2 \end{cases}$$

$$a^{(1)}_{VS} = \begin{cases} \mathbf{J}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} > T_1 \\ \mathbf{O}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} \le T_2 \end{cases} \tag{17}$$

$$a^{(2)}_{HV} = \begin{cases} \mathbf{J}, & T_2 < \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} \le T_1 \\ \mathbf{O}, & T_1 < \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} \le T_2 \end{cases}$$

$$a^{(2)}_{HV} = \begin{cases} \mathbf{J}, & T_2 < \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} \le T_1 \\ \mathbf{O}, & T_1 < \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} \le T_2 \end{cases}$$

$$a^{(2)}_{HV} = \begin{cases} \mathbf{J}, & T_2 < \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} \le T_1 \\ \mathbf{O}, & T_1 < \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} \le T_2 \end{cases} \tag{18}$$

$$a^{(3)}_{HV} = \begin{cases} \mathbf{O}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} > T_2 \\ \mathbf{J}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{HS}}{2} \le T_2 \end{cases}$$

$$a^{(3)}_{HV} = \begin{cases} \mathbf{O} & \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} > T_2 \\ \mathbf{J} & \dfrac{\mathrm{Att}_{HS} + \mathrm{Att}_{VS}}{2} \le T_2 \end{cases}$$

$$a^{(3)}_{HV} = \begin{cases} \mathbf{O}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} > T_2 \\ \mathbf{J}, & \dfrac{\mathrm{Att}_{HV} + \mathrm{Att}_{VS}}{2} \le T_2. \end{cases} \tag{19}$$

Then, the input feature map $m_{i,j}$ is multiplied with three sets of all-zeros and all-ones matrices to obtain the feature-enhanced or penalized matrix f_{HV}

$$\begin{cases} f^{(1)}_{HV} = a^{(1)}_{HV} * m_{i,j} \\ f^{(1)}_{HS} = a^{(1)}_{HS} * m_{i,j} \\ f^{(1)}_{VS} = a^{(1)}_{VS} * m_{i,j} \end{cases} \tag{20}$$

$$\begin{cases} f^{(2)}_{HV} = a^{(2)}_{HV} * m_{i,j} \\ f^{(2)}_{HS} = a^{(2)}_{HS} * m_{i,j} \\ f^{(3)}_{VS} = a^{(3)}_{VS} * m_{i,j} \end{cases} \tag{21}$$

$$\begin{cases} f^{(3)}_{HV} = a^{(3)}_{HV} * m_{i,j} \\ f^{(3)}_{HS} = a^{(3)}_{HS} * m_{i,j} \\ f^{(3)}_{VS} = a^{(3)}_{VS} * m_{i,j} \end{cases} \tag{22}$$

where $*$ denotes multiplication by elements. Next, the output of the CDF module obtained are as follows:

$$f^{(1)}_{\mathrm{cdf}} = \mathrm{concat}\left(f^{(1)}_{HV}, f^{(1)}_{HS}, f^{(1)}_{VS}\right) \tag{23}$$

$$f^{(2)}_{\mathrm{cdf}} = \mathrm{concat}\left(f^{(2)}_{HV}, f^{(2)}_{HS}, f^{(2)}_{VS}\right) \tag{24}$$

$$f^{(3)}_{\mathrm{cdf}} = \mathrm{concat}\left(f^{(3)}_{HV}, f^{(3)}_{HS}, f^{(3)}_{VS}\right) \tag{25}$$

where concat indicates the matrix concatenation operation.

E. Semantic Aware Module

To enhance the efficacy of feature fusion, we developed a SA module that launches shallow features rich in semantic information to deep feature fusion. By taking the square root of the sum of the squares of the weight matrices in the three directions of Att_H, Att_V, and Att_S, we can obtain two new weight matrices G_{HVS} of shallow global semantic features and weight matrix L_{HVS} of deep local semantic features. The motivation is that when the weighted feature points in any direction are larger than a given threshold, they are more discriminative, allowing the model to retain richer semantic information

$$G_{HVS} = \sqrt{(\mathrm{Att}_H)^2 + (\mathrm{Att}_V)^2 + (\mathrm{Att}_S)^2} \tag{26}$$

$$L_{HVS} = \sqrt{(\mathrm{Att}_H)^2 + (\mathrm{Att}_V)^2 + (\mathrm{Att}_S)^2}. \tag{27}$$

Then, SA performs feature screening on shallow global features G_{HVS} and deep local features L_{HVS}. This article first

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

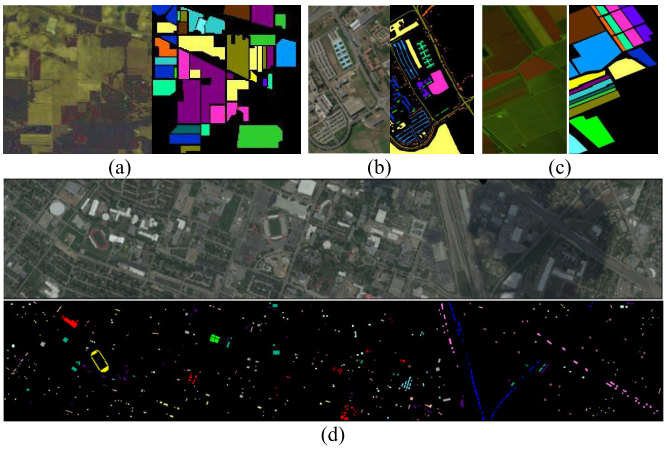


Fig. 4. Ground-truth and false-color composite image for each dataset. (a) IP. (b) PU. (c) SA. (d) Houston.

compares the weight of G_{HVS} with the threshold T_1 to produce the corresponding shallow global matrix $\hat{G}_{HVS}$ with all-zeros and all-ones matrices. Similarly, for the deep local feature L_{HVS}, all-zeros and all-ones matrices of the corresponding deep local matrix $\hat{L}_{HVS}$ can be generated by comparing L_{HVS} with the threshold T3. Specifically, feature screening is the same as the method elucidated earlier, which can be described as follows:

$$\hat{G}^i_{HVS} = \begin{cases} \mathbf{J}(\hat{G}^i_{HVS}), & G_{HVS} > T_1 \\ \mathbf{O}(\hat{G}^i_{HVS}), & G_{HVS} \le T_1 \end{cases} \tag{28}$$

$$\hat{L}^i_{HVS} = \begin{cases} \mathbf{J}(\hat{L}^i_{HVS}), & L_{HVS} > T_3 \\ \mathbf{O}(\hat{L}^i_{HVS}), & L_{HVS} \le T_3. \end{cases} \tag{29}$$

Finally, a set of more discriminative shallow global semantic features f_G and a set of more discriminative deep local semantic features f_L are produced

$$f^i_G = \hat{G}_{HVS} * f^{(i)}_{\text{cdf}} \tag{30}$$

$$f^i_L = \hat{L}_{HVS} * f^{(i)}_{\text{cdf}} \tag{31}$$

where $*$ represents multiplication by elements.

In addition, to investigate the feature representation of redundant features $f^{(3)}_{\text{cdf}}$ of HSIs, we project them into a 1-D vector space to reconstruct feature relations. Then, shallow global features and salient features are fused, and the classifier soft max is applied to classify HSIs.

IV. Experiments

A. Datasets

This article investigates the proposed SACDF on four datasets of HSIs (as shown in Tables I–IV and Fig. 4) involving unbalanced and small samples.

1) The Indian Pines (IP) dataset is HSIs gathered in 1992 using the AVIRIS hyperspectral sensor to image an Indian pine tree in Indiana, United States. It comprises 145 × 145 pixels and 224 spectral reflection bands. After removing the spectral bands encompassing the water-absorbed region, 200 spectral bands remain, and the ground truth (GT) consists of 16 classes; therefore, it is more prone to mixed image elements, which makes classification difficult.

TABLE I
Selected Training Samples for Evaluation From the IP Dataset

ID	Class	Total	5%	10%	15%
C_1	Class_Corn-notill	1428	71	142	214
C_2	Class_Alfalfa	46	2	4	6
C_3	Class_Corn	237	11	23	35
C_4	Class_Corn-mintill	830	41	83	124
C_5	Class_Grass-trees	730	36	73	109
C_6	Class_Grass-pasture	483	24	48	72
C_7	Class_Hay-windrowed	478	23	47	71
C_8	Class_Grass-pasture-mowed	28	1	2	4
C_9	Class_Soybean-notill	972	48	97	145
C_{10}	Class_Oats	20	1	2	3
C_{11}	Class_Soybean-clean	593	29	59	88
C_{12}	Class_Soybean-mintill	2455	122	245	368
C_{13}	Class_Woods	1265	63	126	189
C_{14}	Class_Wheat	205	10	20	30
C_{15}	Class_Stone-Steel-Towers	93	4	9	13
C_{16}	Class_Buildings-Grass-Trees-Drives	386	19	38	57

TABLE II
Selected Training Samples for Evaluation From the PU Dataset

ID	Class	Total	1%	5%	10%
C_1	Class_Meadows	18649	186	932	1864
C_2	Class_Asphalt	6631	66	331	633
C_3	Class_ Trees	3064	30	153	306
C_4	Class_Gravel	2099	20	104	209
C_5	Class_Bare Soil	5029	50	251	502
C_6	Class_Painted metal sheets	1345	13	67	134
C_7	Class_Shadows	947	9	47	94
C_8	Class_Self-Blocking Bricks	3682	36	184	368
C_9	Class_Bitumen	1330	13	66	133

TABLE III
Selected Training Samples for Evaluation From the SA Dataset

ID	Class	Total	1%	5%	10%
C_1	Class_Brocoli_green_weeds_2	3726	37	186	372
C_2	Class_Brocoli_green_w eeds_1	2009	20	100	200
C_3	Class_Fallow_rough_plow	1394	13	69	139
C_4	Class_Fallow	1976	19	98	197
C_5	Class_Stubble	3959	39	197	395
C_6	Class_Fallow_smooth	2678	26	133	267
C_7	Class_Grapes_untrained	11271	112	563	1127
C_8	Class_Celery	3579	35	178	357
C_9	Class_Corn_senesced_green_w eeds	3278	32	163	327
C_{10}	Class_Soil_vinyard_develop	6203	62	310	620
C_{11}	Class_Lettuce_romaine_5wk	1927	19	96	192
C_{12}	Class_Lettuce_romaine_4wk	1068	10	53	106
C_{13}	Class_Lettuce_romaine_7wk	1070	10	53	107
C_{14}	Class_Lettuce_romaine_6wk	916	9	45	91
C_{15}	Class_Vinyard_vertical_trellis	1807	18	90	180
C_{16}	Class_Vinyard_untrained	7268	72	363	726

2) The Pavia University (PU) dataset is a portion of the HSIs dataset imaged by the German ROSIS hyperspectral sensor in 2003 in the city of Pavia, Italy. The sensor has a number of 115 bands, 12 of which were removed due to noise, resulting in 103 bands of usable data with a spatial resolution of 1.3 m. The GT contains nine categories. The dataset is characterized by complex

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SELECTED TRAINING SAMPLES FOR EVALUATION FROM THE HOUSTON DATASET

ID	Class	Total	0.1%	0.5%	1%
C_1	Class_Stressed grass	1254	13	63	125
C_2	Class_Healthy grass	1251	13	63	125
C_3	Class_Trees	1244	12	62	124
C_4	Synthetic grass	697	7	35	70
C_5	Class_Water	325	3	16	33
C_6	Class_Soil	1242	12	62	124
C_7	Class_Commercial	1244	12	62	124
C_8	Class_Residential	1268	13	63	127
C_9	Class_Highway	1227	12	61	123
C_{10}	Class_Road	1252	13	63	125
C_{11}	Class_Parking Lot1	1233	12	62	123
C_{12}	Class_Railway	1235	12	62	124
C_{13}	Class_Tennis court	428	4	21	43
C_{14}	Class_Running track	660	7	33	66
C_{15}	Class_Parking Lot2	469	5	23	47

background pixels, a small connected area, and the presence of a large number of discontinuous pixels.

3) The Salinas (SA) dataset was also imaged by the AVIRIS hyperspectral sensor for the Salinas Valley in California, United States. This dataset is less difficult to classify compared to the other three HSI datasets.
4) The Houston dataset was gathered by the CASI-1500 hyperspectral imaging sensor over the University of Houston and its surrounding regions in 2013 with a spatial resolution of 2.5 m, a spectral wavelength range of 380–1050 nm, and 144 bands, making it a large and complex hyperspectral dataset.

B. Experimental Setup

In this work, we propose to evaluate the proposed SACDF using 0.1%, 1%, 5%, 10%, and 15% training samples on four well-known HSI data. The nine methods we compared are divided into those based on unused attention: SSUN [29], MCNN-CP [30], Tri-CNN [31], MDGCN [32], Gia-CFSL [33] and those based on used attention: MDN-ASP [34], RIAN [35], DMuCA [36], and WFCG [37]. For a fair comparison, these nine algorithms have the same parameter settings and training environment as the proposed SACDF, and to reduce the fluctuation of classification results caused by the randomness of sample selection, the experiments were repeated ten times, and the average value of these experiments is used as the final result. In addition, the total number of SACDF parameters is 375 240, the batch size is 64, and the number of iterations is 128.

C. Evaluation Indicators

To evaluate the robustness of all algorithms in a comprehensive manner, three commonly used metrics, such as overall accuracy (OA), which assesses the ratio of correctly classified pixels; kappa coefficient (Kappa), which measures the classification effectiveness by a series of criteria, and average accuracy (AA), which assesses the ratio of correctly classified pixels in each category of features, are used in the experiments to record and evaluate the classification ability of all algorithms

$$f_{\mathrm{OA}} = \frac{1}{N}\sum_{i=1}^{K} p_{ii} \tag{32}$$

where K and N indicates the amount of object classes and the full amount of pixels, and p_{ii} symbolizes the element at coordinate (i, i) in the confusion matrix.

AA is calculated by calculating the ratio between the correct prediction of each class and the pixels belonging to this class in the real situation and then taking the average value of the accuracy of each class. The calculation equation is as follows:

$$f_{\mathrm{AA}} = \frac{1}{K}\sum_{i=1}^{K} \frac{p_{ii}}{\sum_{j=1}^{K} p_{ij}} \tag{33}$$

where $\sum_{j=1}^{K} p_{ij}$ represents the sum of the ith row elements.

The calculation of the Kappa coefficient can be expressed as follows:

$$f_{\mathrm{Kappa}} = \frac{p_0 - p_e}{1 - p_e} \tag{34}$$

where $p_0 = f_{\mathrm{OA}}$, and p_e can be defined as follows:

$$p_e = \sum_{i=1}^{K} \frac{\sum_{j=1}^{K} p_{ij} \cdot \sum_{j=1}^{K} p_{ji}}{N \times N} \tag{35}$$

where $\sum_{j=1}^{K} p_{ji}$ represents the sum of elements in column i.

TABLE V
COMPARISON RESULTS OF THE RUNNING TIMES OF DIFFERENT METHODS ON THE IP DATASET

Type	IP (5%)	
	Training Time (s)	Test Time (s)
SSUN(2018) [29]	102.69	3.57
MCNN-CP(2020) [30]	571.26	5.80
Tri-CNN(2023) [31]	539.22	4.69
MDN-ASP(2021) [32]	720.16	21.12
RIAN(2022) [33]	43.51	3.16
DMuCA(2022) [34]	64.17	2.82
MDGCN(2019) [35]	93.00	3.36
Gia-CFSL(2022) [36]	6572.23	52.63
WFCG(2022) [37]	42.68	2.66
SACDF	**41.55**	**2.62**

D. Comparison Results With Different SOTA Models

Table V first reports the training and testing times of the different methods on the IP dataset with a training sample size of 5%. The information of all the devices involved in the experiments is as follows: the CPU is an AMD Ryzen 1700X, the RAM is 48 GB, and the GPU is a GTX 1080 (8G). The proposed method clearly has the least time consumption in the training and testing phases, and also, the time consumption of the attention-based RIAN, DMuCA, MDGCN, and WFCG is significantly less than that of the methods, such as SSUN, MCNN-CP, and Gia-CFSL, which do not use attention, and thus, the proposed attention-based SACDF is satisfactory for the computational consumption requirements of edge devices.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON IP

Models	5%			10%			15%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018)[29]	0.7379	0.7359	0.7006	0.7782	0.7343	0.7459	0.8563	0.8663	0.8439
MCNN-CP(2020)[30]	0.7769	0.7778	0.7455	0.8356	0.8158	0.8126	0.8596	0.8495	0.8404
Tri-CNN(2023)[31]	0.8926	0.9036	0.8866	0.9146	0.9069	0.9034	0.9245	0.9169	0.9258
MDN-ASP(2021)[32]	0.9038	0.9154	0.9122	0.9274	0.9245	0.9144	0.9425	0.9397	0.9547
RIAN(2022)[33]	0.9199	0.9187	0.9131	0.9285	0.9301	0.9241	0.9431	0.9411	0.9550
DMuCA(2022)[34]	0.9233	0.9310	0.9289	0.9693	0.9597	0.9650	0.9569	0.9608	0.9587
MDGCN(2019)[35]	0.9236	0.9318	0.9068	0.9686	0.9580	0.9642	0.9726	0.9679	0.9739
Gia-CFSL(2022)[36]	0.9517	0.9389	0.9479	0.9715	0.9602	0.9734	0.9827	0.9811	0.9767
WFCG(2022)[37]	0.9604	0.9429	0.9496	0.9786	0.9651	0.9799	0.9903	0.9815	0.9837
SACDF	**0.9645**	**0.9521**	**0.9596**	**0.9853**	**0.9806**	**0.9832**	**0.9914**	**0.9844**	**0.9902**

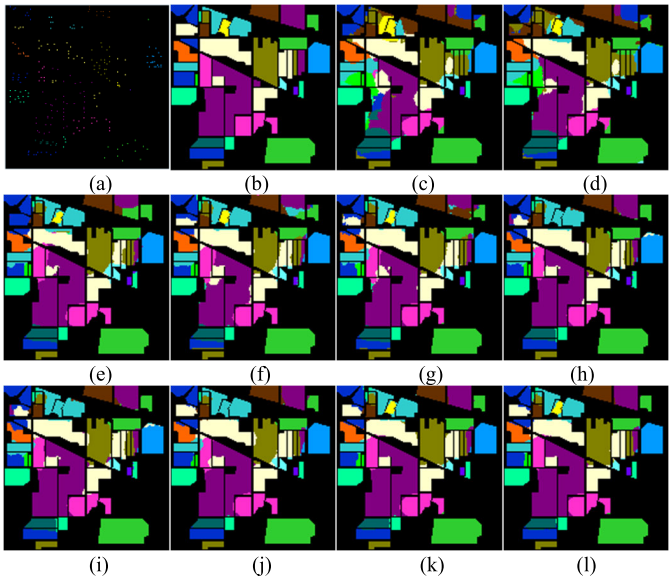


Fig. 5. Visualization results of comparative experiments on IP. (a) Input (5%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) SACDF.

1) Comparison Results on the IP Dataset: It is imbalanced, and training HSIs classifier with minimal training examples presents issues. Since the IP dataset's Oats and Grass-pasture-mowed categories only contain 28 and 20 samples, respectively. In order to train the classifier, we were required to select 5% training samples; the classification results for 10% and 15% training samples are also presented in this study. According to Table VI and Fig. 5, SSUN has the weakest classification performance and the most noise. This is due to the fact that it is a shallow neural network with insufficient generalization ability on limited sample data. Among the remaining eight methods, RIAN and MDN-ASP, also based on CNN, outperformed SSUN, MCNN-CP, and Tri-CNN, which did not use the attention mechanism, indicating a clear advantage of attention for HSI data. Also, the WFCG model using graph learning outperforms Gia-CFSL, which also uses graph learning due to the introduction of the attention mechanism. In addition, the classification accuracy achieved using the attention model is much higher than that of SSUN and MCNN-CP. This is because the trained attention mechanism tends to focus more on differentiating features. With the three-direction attention and SA modules, SACDF achieves the best classification performance. Notably, MDGCN with inferior classification performance beats Tri-CNN due to the attention mechanism drive, suggesting that the attention module can potentially improve the robustness of the HSIs classifier. It is, therefore, clear that the superiority of SACDF is demonstrated.

Among the remaining eight methods, MDN-ASP, RIAN, and DMuCA, which used the attention mechanism, achieved significantly better classification performance than SSUN, MCNN-CP, and Tri-CNN, which did not use the attention mechanism, indicating the superiority of attention on the HSI classification task. In addition, although these three CNN-based attention methods were slightly inferior to the graph learning-based MDGCN and Gia-CFSL, the WFCG that used attention achieved superior classification capability over MDGCN and Gia-CFSL in the same situation of using graph learning. Thus, the attention mechanism helps to focus on more discriminative features in hyperspectral data. Finally, this is further supported by the best classification results achieved by the SACDF using stereo attention, with improvements of 0.43%–23.49%, 0.15%–22.70%, and 1.04%–26.99% over the compared methods in the three metrics OA, AA, and Kappa.

2) Comparison Results on PU Datasets: Complex backdrop pixels, a small number of connected regions, and a big number of discontinuous pixels define the PU dataset. The experimental comparison findings indicate that the performance of each technique on this dataset is slightly enhanced. This may be because there are only nine classes in this dataset, and the spatial resolution is great, which decreases the classification complexity slightly. In the event of a very limited number of labeled data, the performance of the baseline technique remains inadequate; however, the proposed SACDF model achieved a competitive performance, with the classification results achieving good classification results for discontinuous types of features compared to other algorithms. It can also be found from Table VII and Fig. 6 that RIAN and DMuCA using attention mechanism on AA outperform MDGCN and Gia-CFSL using graph learning, respectively, indicating that the attention mechanism has stronger feature mining ability in small sample scenarios. Similarly, the superiority of WFCG using attention and graph learning over MDGCN and Gia-CFSL using only graph learning also illustrates the superiority

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON PU

Models	1%			5%			10%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018)[29]	0.8546	0.8768	0.8428	0.9359	0.9247	0.9151	0.9482	0.9362	0.9314
MCNN-CP(2020)[30]	0.8599	0.8205	0.8116	0.9395	0.9251	0.9197	0.9482	0.9413	0.9314
Tri-CNN(2023)[31]	0.9278	0.9081	0.9003	0.9413	0.9455	0.9537	0.9568	0.9627	0.9545
MDN-ASP(2021)[32]	0.9294	0.9082	0.9119	0.9538	0.9527	0.9414	0.9569	0.9678	0.9601
RIAN(2022)[33]	0.9358	0.9183	0.9248	0.9551	0.9602	0.9433	0.9608	0.9682	0.9614
DMuCA(2022)[34]	0.9364	0.9204	0.9219	0.9797	0.9751	0.9801	0.9709	0.9721	0.9633
MDGCN(2019)[35]	0.9378	0.9158	0.9267	0.9682	0.9517	0.9633	0.9768	0.9628	0.9712
Gia-CFSL(2022)[36]	0.9357	0.9214	0.9317	0.9725	0.9628	0.9733	0.9822	0.9725	0.9844
WFCG(2022)[37]	0.9456	0.9301	0.9345	0.9855	0.9729	0.9817	0.9934	0.9871	0.9914
SACDF	**0.9500**	**0.9315**	**0.9395**	**0.9878**	**0.9783**	**0.9839**	**0.9941**	**0.9875**	**0.9915**

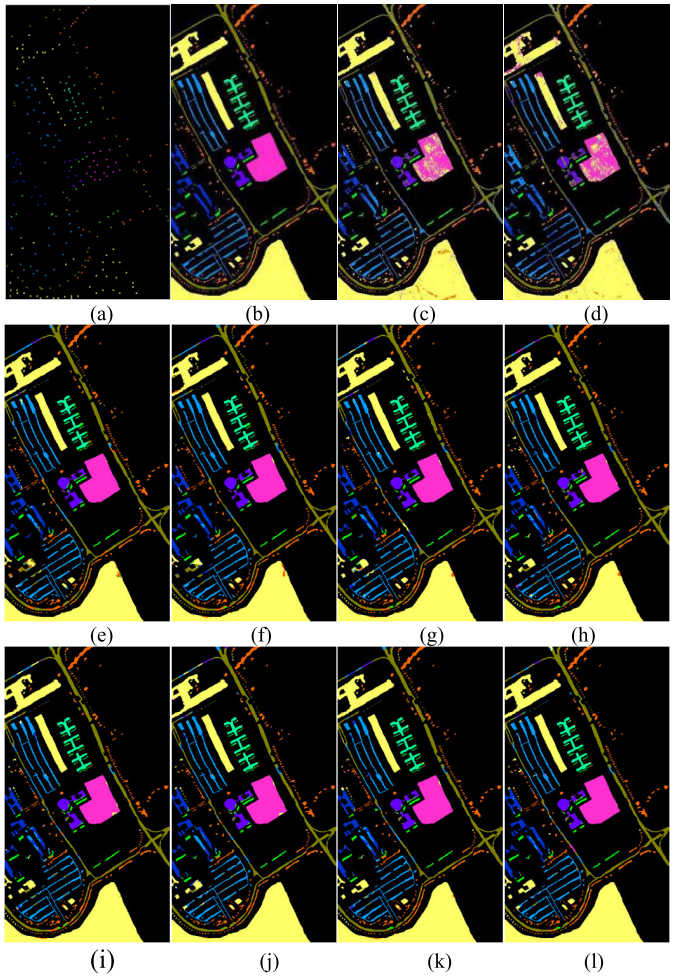


Fig. 6. Visualization results of comparative experiments on PU. (a) Input (1%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) SACDF.

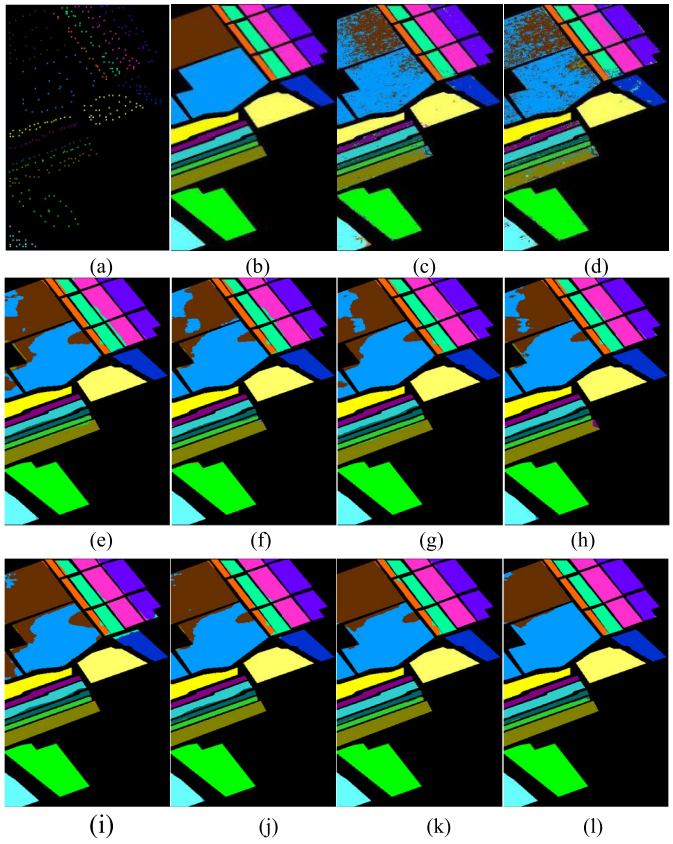


Fig. 7. Visualization results of comparative experiments on SA. (a) Input (1%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) SACDF.

of attention. Finally, the proposed SACDF provides state-of-the-art capabilities on all three metrics, with OA, AA, and Kappa improving by 0.46%–10.04%, 0.15%–11.91%, and 0.53%–13.61%, respectively, at 1% of the training sample.

3) Comparison Results on the SA Dataset: As can be seen from Table VIII and Fig. 7, all classifiers achieved high OA on SA, but on the Grapes_untrained class, the approach using attention outperformed the approach using only CNN or graph learning, the Grapes_untrained class is spatially discontinuous and the attention mechanism clearly tapped into this better, indicating that that attention effectively obtains spatial dependencies between spectra; furthermore, the designed SACDF method improved the three metrics by 0.01%–9.49%, 0.65%–9.66%, and 0.56%–11.51%, respectively.

4) Comparison Results on the Houston Dataset: The Houston 2013 is larger and more complex than the three preceding datasets; therefore, we used only 0.1% of the samples for training. As shown in Table IX and Fig. 8, all methods produced high OA on Houston, but the proposed SACDF remained the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON SA

Models	1%			5%			10%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018)[29]	0.8959	0.9334	0.8841	0.9033	0.8823	0.8922	0.9195	0.8944	0.9102
MCNN-CP(2020)[30]	0.8862	0.9177	0.8731	0.9211	0.9462	0.9121	0.9369	0.9594	0.9286
Tri-CNN(2023)[31]	0.9271	0.8927	0.9043	0.9322	0.9288	0.9311	0.9536	0.9617	0.9526
MDN-ASP(2021)[32]	0.9366	0.9217	0.9204	0.9444	0.9547	0.9427	0.9552	0.9756	0.9625
RIAN(2022)[33]	0.9397	0.9254	0.9305	0.9546	0.9601	0.9488	0.9634	0.9759	0.9688
DMuCA(2022)[34]	0.9452	0.9367	0.9401	0.9587	0.9725	0.9614	0.9715	0.9801	0.9714
MDGCN(2019)[35]	0.9462	0.9498	0.9512	0.9673	0.9822	0.9636	0.9868	0.9817	0.9725
Gia-CFSL(2022)[36]	0.9662	0.9758	0.9699	0.9858	0.9816	0.9825	0.9963	0.9924	0.9826
WFCG(2022)[37]	0.9799	0.9817	0.9811	0.9895	0.9934	0.9914	0.9984	0.9988	0.9971
SACDF	**0.9880**	**0.9882**	**0.9867**	**0.9990**	**0.9982**	**0.9991**	**0.9996**	**0.9993**	**0.9995**

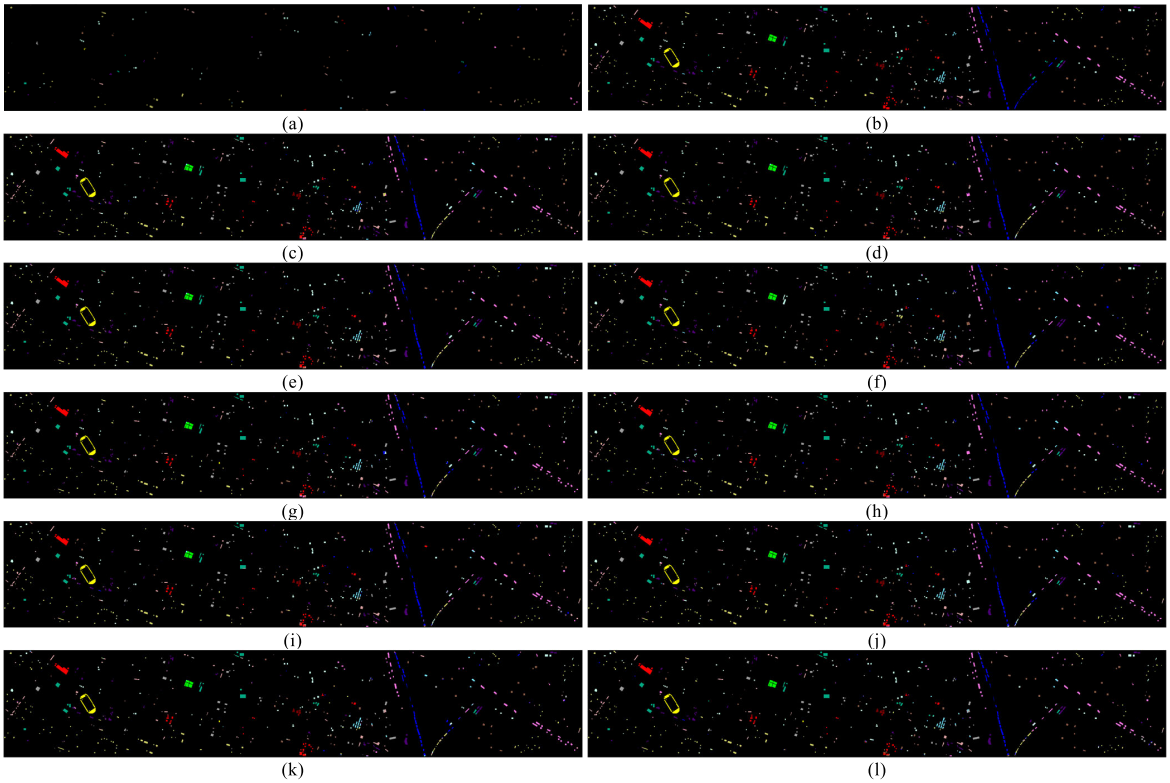


Fig. 8. Visualization results of comparative experiments on Houston. (a) Input (5%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) SACDF.

best by assigning weights to each feature to extract attentional features, decomposing shallow features in three directions—horizontal, vertical, and spatial—and introducing a weight distribution between spatial attentional feature maps to further obtain spatial features for optimal classification performance, thus improving the classifier's ability to perceive features with discriminative power. The OA, AA, and Kappa coefficients of SACDF at 0.1% training sample were 0.34%–3.68%, 0.36%–5.67%, and 1.01%–13.61% higher than those of other methods, respectively.

E. Experimental Results of Federated Neural Learning

Hyperspectral image classification technology is often used in various monitoring and military tasks and requires high data security and reliability; therefore, this section conducts experimental demonstrations from the aspects of safety and reliability of hyperspectral classification and computational consumption. First of all, this article simulates an edge computing application scenario with one client device, one security aggregator, and one master server on a computer with AMD Ryzen 7 1700X CPU, 48 GB memory, and Windows 10 operating system. Each client device has the same computing power, and a dataset is sent to each client device by category; the experimental results are shown in Table X below.

It can be clearly seen from Table X and Fig. 9 that after the dataset is divided into k parts according to categories, they have the same feature space, so the privacy of hyperspectral data is greatly improved, even if a single or several customers. If the end device is attacked or destroyed, as long as there is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON HOUSTON

Models	0.1%			0.5%			1%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018)[29]	0.8125	0.8056	0.8231	0.9435	0.9520	0.9389	0.9845	0.9755	0.9829
MCNN-CP(2020)[30]	0.8269	0.8397	0.8178	0.9582	0.9627	0.9401	0.9817	0.9623	0.9779
Tri-CNN(2023)[31]	0.8296	0.8229	0.8355	0.9528	0.9436	0.9418	0.9625	0.9601	0.9734
MDN-ASP(2021)[32]	0.8299	0.8309	0.8367	0.9587	0.9527	0.9484	0.9719	0.9882	0.9752
RIAN(2022)[33]	0.8305	0.8311	0.8297	0.9627	0.9714	0.9732	0.9787	0.9799	0.9827
DMuCA(2022)[34]	0.8314	0.8323	0.8257	0.9605	0.9710	0.9741	0.9810	0.9757	0.9846
MDGCN(2019)[35]	0.8397	0.8257	0.8225	0.9617	0.9526	0.9596	0.9861	0.9859	0.9812
Gia-CFSL(2022)[36]	0.8405	0.8472	0.8322	0.9714	0.9703	0.9759	0.9901	0.9825	0.9834
WFCG(2022)[37]	0.8407	0.8510	0.8303	0.9643	0.9712	0.9628	0.9899	0.9903	0.9917
SACDF	**0.8436**	**0.8541**	**0.8387**	**0.9755**	**0.9741**	**0.9773**	**0.9905**	**0.9907**	**0.9922**

TABLE X
EXPERIMENTAL RESULTS OF FEDERATED LEARNING ON PU DATASET

	(k client devices, 1 security aggregator, 1 master server)						
	k=1	k=2	k=3	k=4	k=5	k=10	k=20
OA	0.9999	0.9998	0.9988	0.9966	0.9943	0.9941	0.9878
AA	0.9999	0.9999	0.9957	0.9959	0.9895	0.9875	0.9783
Kappa	0.9999	0.9997	0.9986	0.9961	0.9936	0.9915	0.9839

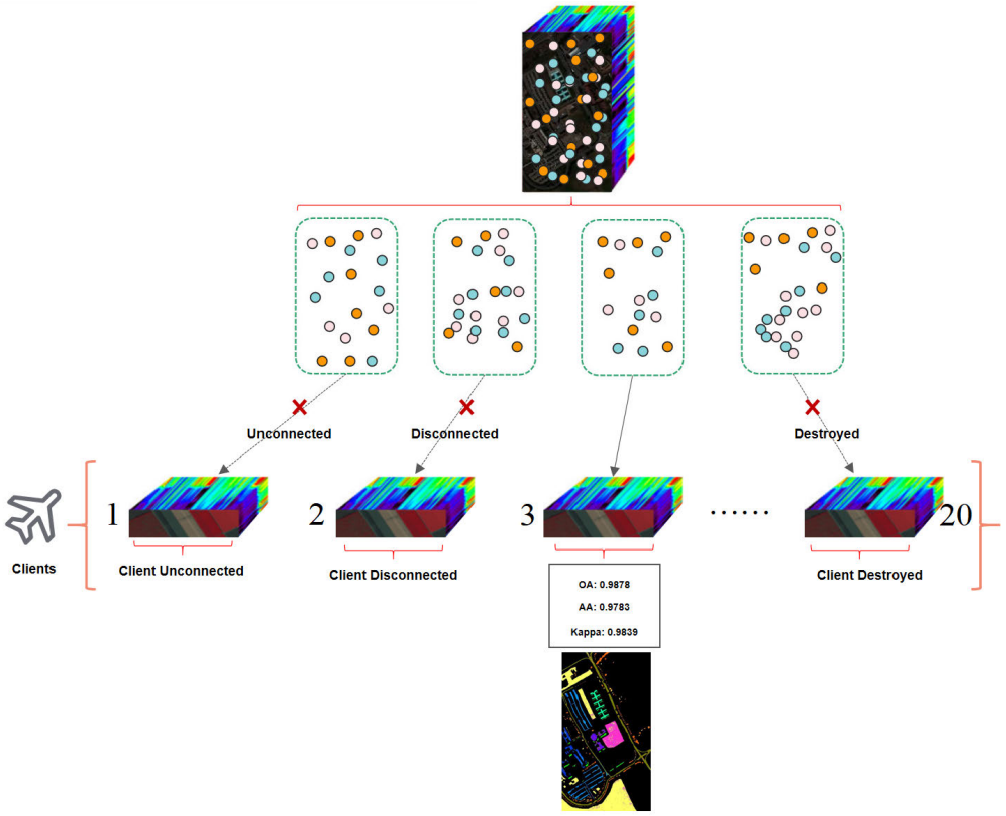


Fig. 9. Schematic of federated neural learning-based high-reliability classification of hyperspectral images.

one client device online, the entire task will not be affected, and the reliability of the HSI classification system will be greatly improved. In addition, it can also be found that when using 20 devices for federated learning, the accuracy rate is significantly lower when using only ten devices. Since the OA rate of 20 devices is as high as 98.78%, it is reasonable to set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

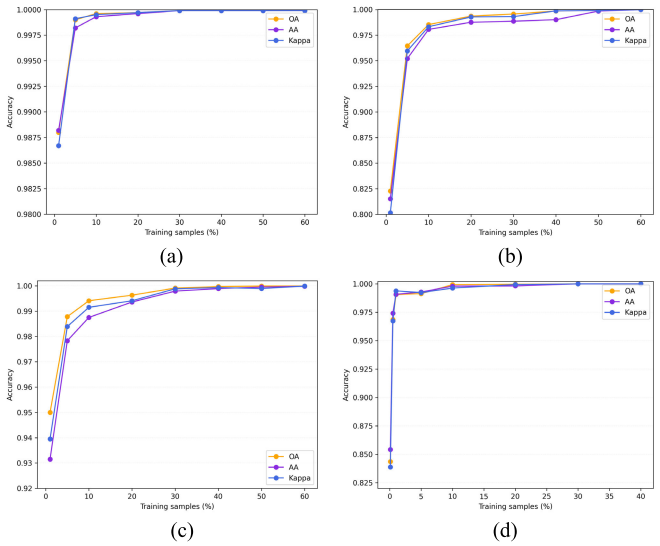


Fig. 10. Performance trends of the SACDF with increasing training samples. (a) SA. (b) IP. (c) PU. (d) Houston.

k to 20 in the current situation, and it has high reliability when applied to military tasks. This also proves that the HSI classification algorithm based on improved federated neural learning proposed in this article is effective.

F. Performance Trends of Different Training Samples

This section focuses on training robust classifiers on few samples, but to thoroughly test the efficacy of the designed SACDF, it uses 0.1%–60% samples to train on the IP, and the test data are displayed in Fig. 10. Despite the fact that only 5% of the training data is used by the proposed method on IP, the OA is still as high as 96.04%. The performance curves on the four datasets clearly illustrate the sharp increase in classification ability as the number of training samples grows, and all obtain close to 100% OA when using 30% of the samples, which fully demonstrates the superiority of SACDF in a limited sample environment.

G. Ablation Experiments for Federated Neural Learning

Since one of the main contributions of the proposed method is the use of federated learning techniques to enhance the privacy and reliability of HSI classification and recognition tasks, specifically, instead of randomly slicing and dicing the sample data to be distributed to the client devices, we divide the datasets by category label so that they have the same feature space; therefore, to demonstrate the superiority of this method, we conduct ablation experiments. "Random" represents the random division of the dataset, and "Category Label" represents the division of the dataset by category labels. The experimental results are shown in Table XI.

From Table XI, it can be seen that the classification results of seven random divisions of the data are highly uncertain and poorly classified, whereas good classification results can be achieved by category labeling, which further proves the superiority of the proposed federated learning method; however, its limitation is that division by category labeling requires the labels of the dataset to be determined in advance, and therefore, it is currently not applicable to unsupervised learning.

TABLE XI
ABLATION EXPERIMENT RESULTS OF FEDERATED NEURAL LEARNING

Type	IP(5%)		
	OA	AA	Kappa
Random (1)	0.8025	0.7754	0.7757
Random (2)	0.9573	0.9374	0.9514
Random (3)	0.8715	0.8513	0.8637
Random (4)	0.8153	0.7799	0.7889
Random (5)	0.8915	0.8728	0.8861
Random (6)	0.8208	0.7849	0.7952
Random (7)	0.9158	0.9079	0.9126
Category Label (Ours)	**0.9645**	**0.9521**	**0.9596**

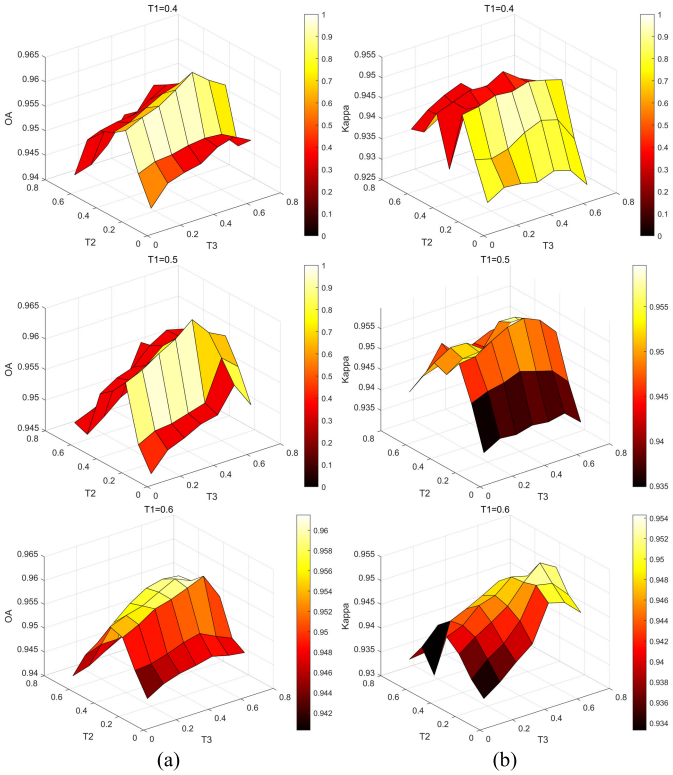


Fig. 11. Classification results with gating thresholds for different T_1, T_2, and T_3 on IP. (a) OA. (b) Kappa.

H. Ablation Experiments for Gated Threshold

In this section, a gated threshold ablation experiment is conducted in order to verify the efficacy of the gated threshold screening strategy. Table XII and Fig. 11 show the modulo output results at different gating thresholds. In the IP dataset, when T_2 and T_3 are fixed, as the gating threshold T_1 increases, the classification performance improves and then gradually decreases. As T_2 increases, the classification performance shows the same trend when T_1 and T_3 are fixed. The reason may be that when the gating threshold is low, irrelevant, redundant information cannot be effectively filtered, thereby diminishing the classification performance of the model as a whole. In addition, when the gating threshold is raised to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
ABLATION EXPERIMENT RESULTS OF GATING THRESHOLD

Threshold	IP(5%)		
	OA	AA	Kappa
T1 = 0.5, T2 = 0.3, T3 = 0.5	**0.9645**	**0.9521**	**0.9596**
T1 = 0.6, T2 = 0.3, T3 = 0.6	0.9615	0.9508	0.9543
T1 = 0.6, T2 = 0.2, T3 = 0.5	0.9520	0.9423	0.9521
T1 = 0.6, T2 = 0.2, T3 = 0.4	0.9512	0.9436	0.9457
T1 = 0.6, T2 = 0.4, T3 = 0.5	0.9606	0.9499	0.9510
T1 = 0.7, T2 = 0.5, T3 = 0.5	0.9506	0.9513	0.9533

TABLE XIII
ABLATION EXPERIMENT RESULTS OF SUB-MODULES

Model	Houston(0.1%)		
	OA	AA	Kappa
NO-CDF	0.8132	0.8287	0.8212
NO-SA	0.8255	0.8369	0.8261
Ours	**0.8436**	**0.8541**	**0.8387**

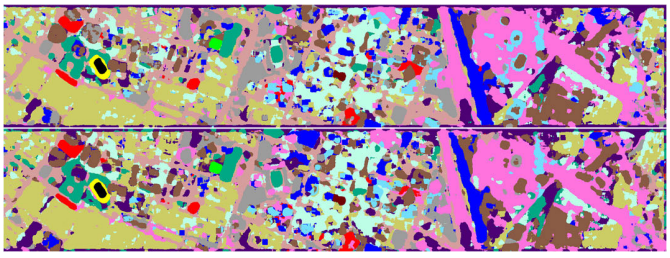


Fig. 12. From top to bottom are the full classification results using 0.1% of the training samples without CDF and without SA, respectively.

a certain peak value, irrelevant, redundant information can be effectively filtered out while spatial semantic details are maintained.

In addition, as the gating threshold rises, some useful features may be filtered out, resulting in insufficient feature representation. From Table XII, it can be seen that the classification performance of the proposed algorithm is the best when T1 = 0.5, T2 = 0.3, T3 = 0.5.

I. Ablation Experiments for Submodules

In order to explore the superiority of the CDF module and the SA module, submodule ablation experiments are conducted in this section, and the experimental results are shown in Table XIII.

As shown in Table XIII and Fig. 12, the classification results without SA outperformed those without CDF, suggesting that the three directions of attention focus on the more discriminative features, providing the basis for SA. Thus, the directions proposed in this article are effective for obtaining stereoscopic attention weights from all three directions.

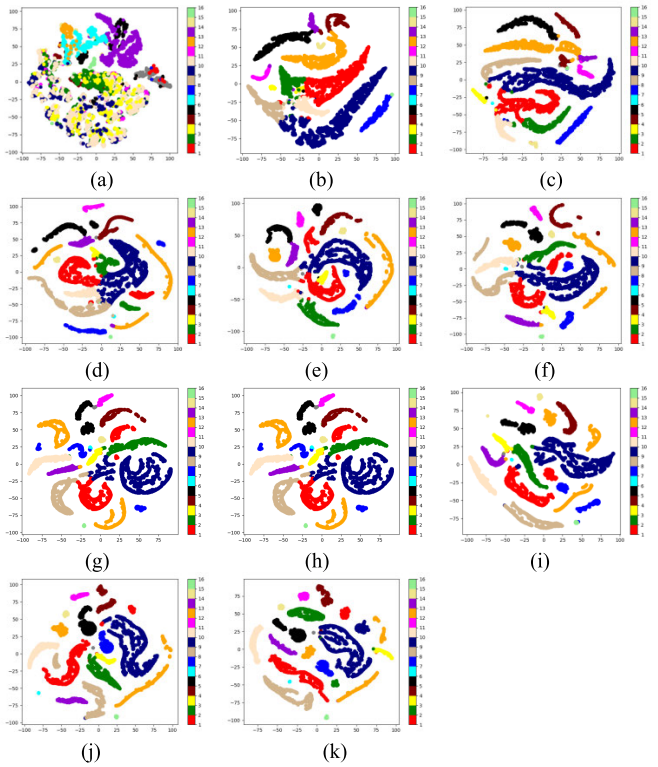


Fig. 13. t-SNE feature visualization of SACDF and the other nine comparison models. (a) Raw. (b) SSUN. (c) MCNN-CP. (d) Tri-CNN. (e) MDN-ASP. (f) RIAN. (g) DMuCA. (h) MDGCN. (i) Gia-CFSL. (j) WFCG. (k) SACDF.

The results reported in Table XIII show that the classification performance decreases by 3.63%, 2.97%, and 2.08% in the three metrics, respectively, after removing the CDF module; however, the classification performance decreases by 2.15%, 2.01%, and 1.50%, respectively, when the SA module is removed. This also demonstrates that the CDF module outperforms the SA module. By assigning various attention weights in different directions, it is possible to get more representative features, which can considerably improve the classification performance of HSIs.

J. t-SNE Data Distributions Visualization

To further confirm the superiority of the proposed method, we visualize the predicted output of the model using t-SNE on the IP dataset. The visualization results of SACDF compared with the other nine sets of classifiers are shown in Fig. 13, from which we can clearly see that the proposed SACDF achieves the best classification performance. In addition, SSUN, MCNN-CP, Tri-CNN, and MDGCN have lower clustering accuracy, while RIAN, DMuCA, and WFCG have clearly better results with more compact intraclass and more interclass separation. This is because stereo attention mines more discriminative features. Of course, the superiority of SACDF is mainly attributed to the proposed stereo attention and CDF modules.

V. CONCLUSION

This article presents a stereo attention CDF-guided federated learning algorithm for hyperspectral image classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

The core idea is to propose a novel federated neural learning algorithm that distributes training samples in the same feature space to distributed client devices by chunking the sample data to improve data privacy and recognition system reliability, as demonstrated in this article through simulation experiments; however, its potential limitation is that the dataset is already labeled, and for the unlabeled dataset it needs to be clustered first using an unsupervised algorithm, and is therefore, limited by the performance of clustering. This article then proposes a novel stereo attention CDF module for hyperspectral image classification, which is able to exploit the multidimensional spatial details of hyperspectral images. In addition, a three-directional (horizontal, vertical, and spatial directions) stereo attention mechanism is proposed to extract the most discriminative features direction by direction, outperforming traditional attention mechanisms that assign attention weights from a single direction only. We demonstrate the superiority of the CDF and SA modules through extensive comparison and ablation experiments, which are able to model multidimensional spatial relationships, and thus, effectively enhance feature representation. Extensive experiments on multiple open-source baseline datasets show that the algorithms in this article provide competitive performance, with the most significant contribution being that the proposed algorithms can effectively enhance the privacy and reliability of hyperspectral image classification and recognition systems.

In future work, we will focus on improving the performance of unsupervised clustering algorithms on HSIs to address the current limitations of working on unlabeled datasets.

REFERENCES

[1] W. Zheng, L. Yin, X. Chen, Z. Ma, S. Liu, and B. Yang, "Knowledge base graph embedding module design for visual question answering model," *Pattern Recognit.*, vol. 120, Dec. 2021, Art. no. 108153.

[2] L. M. Dang, K. Min, H. Wang, M. J. Piran, C. H. Lee, and H. Moon, "Sensor-based and vision-based human activity recognition: A comprehensive survey," *Pattern Recognit.*, vol. 108, Dec. 2020, Art. no. 107561.

[3] L. Greco, G. Percannella, P. Ritrovato, F. Tortorella, and M. Vento, "Trends in IoT based solutions for health care: Moving AI to the edge," *Pattern Recognit. Lett.*, vol. 135, pp. 346–353, Jul. 2020.

[4] S. Wan, S. Ding, and C. Chen, "Edge computing enabled video segmentation for real-time traffic monitoring in Internet of Vehicles," *Pattern Recognit.*, vol. 121, Jan. 2022, Art. no. 108146.

[5] X. Hu, C. Masouros, and K.-K. Wong, "Reconfigurable intelligent surface aided mobile edge computing: From optimization-based to location-only learning-based solutions," *IEEE Trans. Commun.*, vol. 69, no. 6, pp. 3709–3725, Jun. 2021.

[6] G. S. Sriram, "Edge computing vs. cloud computing: An overview of big data challenges and opportunities for large enterprises," *Int. Res. J. Modernization Eng. Technol. Sci.*, vol. 4, no. 1, pp. 1331–1337, 2022.

[7] J. Zhu, X. Lou, and W. Ye, "Lightweight deep learning model in mobile-edge computing for radar-based human activity recognition," *IEEE Internet Things J.*, vol. 8, no. 15, pp. 12350–12359, Aug. 2021.

[8] L. Liu, M. Zhao, M. Yu, M. A. Jan, D. Lan, and A. Taherkordi, "Mobility-aware multi-hop task offloading for autonomous driving in vehicular edge computing and networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 2, pp. 2169–2182, Feb. 2023.

[9] Y. Sun, L. Lei, X. Li, H. Sun, and G. Kuang, "Nonlocal patch similarity based heterogeneous remote sensing change detection," *Pattern Recognit.*, vol. 109, Jan. 2021, Art. no. 107598.

[10] D. H. Svendsen, L. Martino, and G. Camps-Valls, "Active emulation of computer codes with Gaussian processes—Application to remote sensing," *Pattern Recognit.*, vol. 100, Apr. 2020, Art. no. 107103.

[11] A. S. Mohammed, K. Venkatachalam, S. Hubálovský, P. Trojovský, and P. Prabu, "Smart edge computing for 5G/6G satellite IoT for reducing inter transmission delay," *Mobile Netw. Appl.*, vol. 27, no. 3, pp. 1050–1059, Jun. 2022.

[12] S. Purohit and A. K. Jain, "Leveraging IoT and AI for delivering tailor-made customer experiences in Indian petro retailing," *Indian J. Sci. Technol.*, vol. 14, no. 7, pp. 604–609, Feb. 2021.

[13] C. F. Frasser et al., "Fully parallel stochastic computing hardware implementation of convolutional neural networks for edge computing applications," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 22, 2022, doi: 10.1109/TNNLS.2022.3166799.

[14] L. Li, Y. Fan, M. Tse, and K.-Y. Lin, "A review of applications in federated learning," *Comput. Ind. Eng.*, vol. 149, Nov. 2020, Art. no. 106854.

[15] Y. Liu, J. Peng, J. Kang, A. M. Iliyasu, D. Niyato, and A. A. A. El-Latif, "A secure federated learning framework for 5G networks," *IEEE Wireless Commun.*, vol. 27, no. 4, pp. 24–31, Aug. 2020.

[16] B. Gu, A. Xu, Z. Huo, C. Deng, and H. Huang, "Privacy-preserving asynchronous vertical federated learning algorithms for multiparty collaborative learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6103–6115, Nov. 2022.

[17] R. Yu and P. Li, "Toward resource-efficient federated learning in mobile edge computing," *IEEE Netw.*, vol. 35, no. 1, pp. 148–155, Jan. 2021.

[18] J. Xu and H. Wang, "Client selection and bandwidth allocation in wireless federated learning networks: A long-term perspective," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1188–1200, Feb. 2021.

[19] K. Pillutla, S. M. Kakade, and Z. Harchaoui, "Robust aggregation for federated learning," *IEEE Trans. Signal Process.*, vol. 70, pp. 1142–1154, 2022.

[20] Z. Song, H. Sun, H. H. Yang, X. Wang, Y. Zhang, and T. Q. S. Quek, "Reputation-based federated learning for secure wireless networks," *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1212–1226, Jan. 2022.

[21] K. Wei et al., "Federated learning with differential privacy: Algorithms and performance analysis," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3454–3469, 2020.

[22] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020, doi: 10.1109/TGRS.2019.2951433.

[23] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021, doi: 10.1109/TGRS.2020.3015157.

[24] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412, doi: 10.1109/TGRS.2023.3279834.

[25] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021, doi: 10.1109/TCYB.2020.3028931.

[26] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019, doi: 10.1109/TIP.2018.2878958.

[27] O. Ali, M. K. Ishak, M. K. L. Bhatti, I. Khan, and K.-I. Kim, "A comprehensive review of Internet of Things: Technology stack, middlewares, and fog/edge computing interface," *Sensors*, vol. 22, no. 3, p. 995, Jan. 2022.

[28] D.-G. Zhang et al., "A novel edge computing architecture based on adaptive stratified sampling," *Comput. Commun.*, vol. 183, pp. 121–135, Feb. 2022.

[29] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[30] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.

[31] M. Q. Alkhatib et al., "Tri-CNN: A three branch model for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 2, p. 316, Jan. 2023.

[32] B. Xi et al., "Multi-direction networks with attentional spectral prior for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500915.

[33] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022.

[34] M. Liang, Q. He, X. Yu, H. Wang, Z. Meng, and L. Jiao, "A dual multi-head contextual attention network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 13, p. 3091, Jun. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.

[35] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[36] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[37] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022, doi: 10.1109/TIP.2022.3144017.



Weiwei Cai (Member, IEEE) is currently with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. Prior to that, he worked with IT industry for more than ten years in the roles of System Architect and Program Manager. His research interests include machine learning, deep learning, and hyperspectral image processing.

Mr. Cai is an Associate Editor of *Journal of Cloud Computing*. He has also served as a Guest Editor for the *CMES-Computer Modeling in Engineering & Sciences*, *Fire*, and *Crop Protection*. He was ranked among The World's Top 2% of Scientists in 2022 and 2023 by Stanford University.



Ming Gao (Member, IEEE) received the M.Ed. degree in sport medicine from Wuhan Sports University, Wuhan, China, in 2009. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China.

His research interests include neural networks, computer vision, image processing, and hyperspectral image processing.



Yao Ding received the M.S. and Ph.D. degrees from the Key Laboratory of Optical Engineering, Xi'an Research Institute of High Technology, Xi'an, China, in 2013 and 2022, respectively.

His research interests include neural networks, computer vision, image processing, and hyperspectral image clustering. He has published several articles in journals IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), *Information Sciences* (INS), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), and *Neurocomputing*. At present, he has four highly cited papers and two hot paper of ESI.

Dr. Ding is also a reviewer of CMC, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), PR, JAG, and RS.



Xin Ning (Senior Member, IEEE) received the B.S. degree in software engineering from Xinjiang University, Urumqi, China, in 2012 and the Ph.D. degree in electronic circuit and system from the University of Chinese Academy of Sciences, Beijing, China, in 2017.

He is currently an Associate Professor with the Laboratory of Artificial Neural Networks and High Speed Circuits, Institute of Semiconductors, Chinese Academy of Sciences, Beijing. His research interests include pattern recognition, computer vision, and image processing.



Xiao Bai (Senior Member, IEEE) received the B.Eng. degree in computer science from Beihang University, Beijing, China, in 2001, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2006.

He was a Research Officer (Fellow, Scientist) with the Department of Computer Science, University of Bath, bath, U.K., until 2008. He is currently a Full Professor with the State Key Laboratory of Software Development Environment, School of Computer Science and Engineering, Jiangxi Research Institute, Beihang University. He has authored or coauthored more than 100 papers in journals and refereed conferences. His research interests include pattern recognition, image processing, and remote sensing image analysis.

Dr. Bai is an Associate Editor of *Pattern Recognition and Signal Processing*.



Pengjiang Qian (Senior Member, IEEE) received the Ph.D. degree from Jiangnan University, Wuxi, China, in March 2011.

He is currently a Full Professor with the School of Artificial Intelligence and Computer Science, Jiangnan University. He has authored or coauthored more than 80 papers published in international/national journals and conferences, such as IEEE TRANSACTIONS ON MEDICAL IMAGING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS—PART B: CYBERNETICS, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON FUZZY SYSTEMS, *Information Fusion*, *Pattern Recognition*, *Information Sciences*, and *Knowledge-Based Systems*. His research interests include data mining, pattern recognition, bioinformatics, and their applications, such as analysis and processing for medical imaging, intelligent traffic dispatching, and advanced business intelligence in logistics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:34:05 UTC from IEEE Xplore. Restrictions apply.